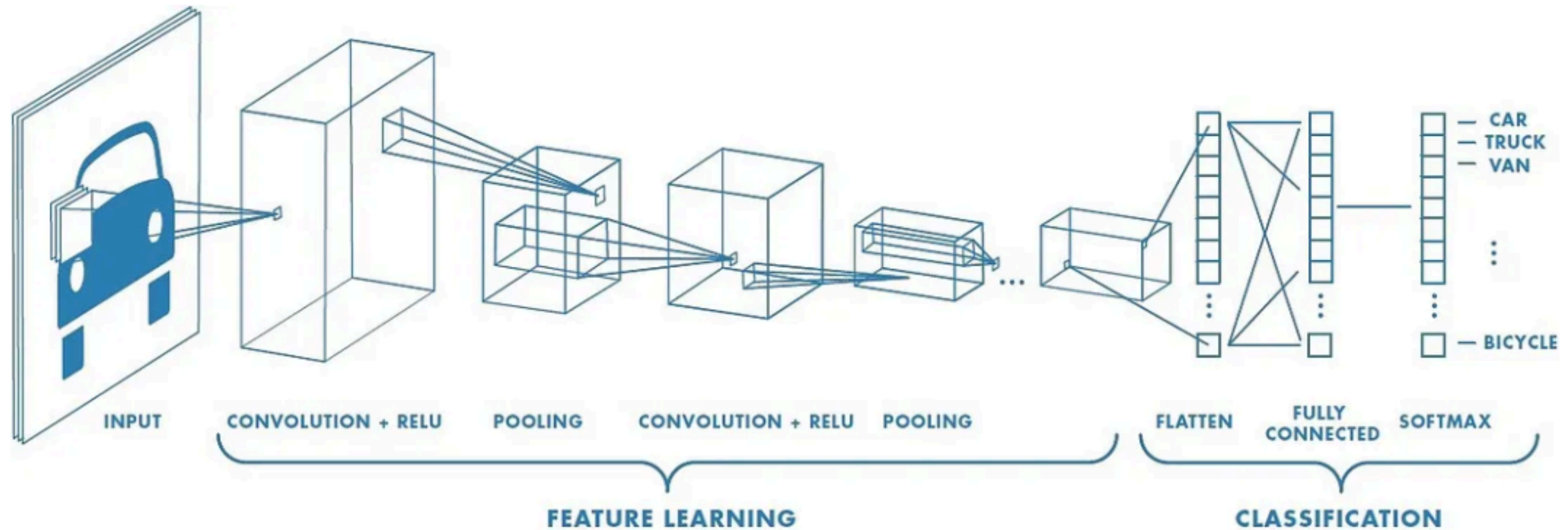
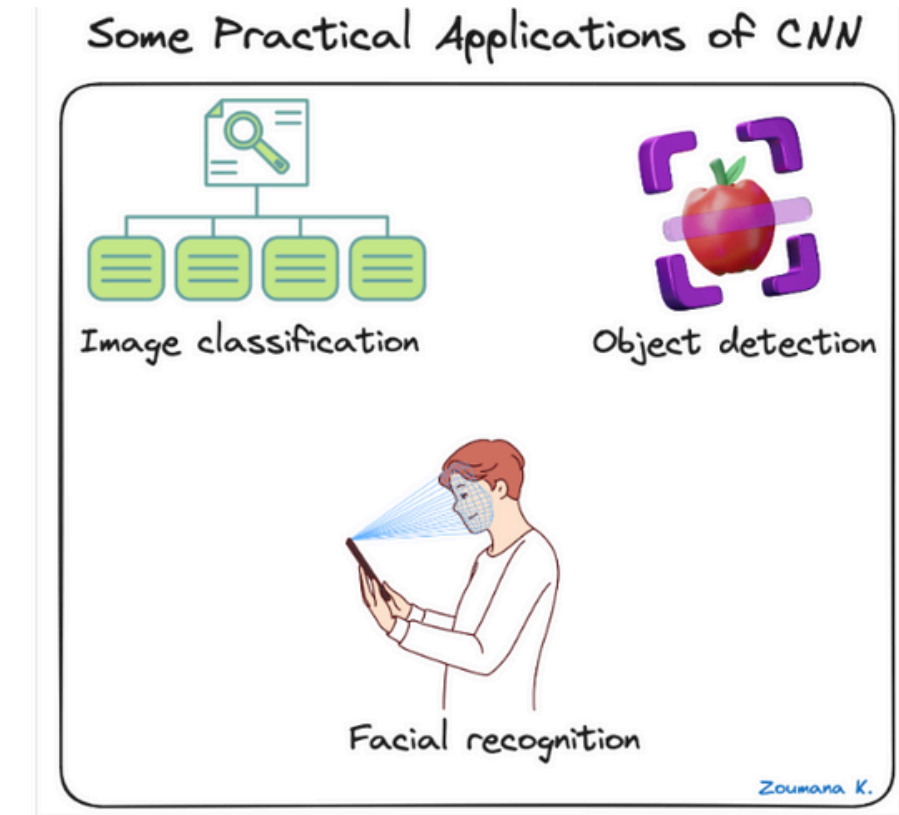
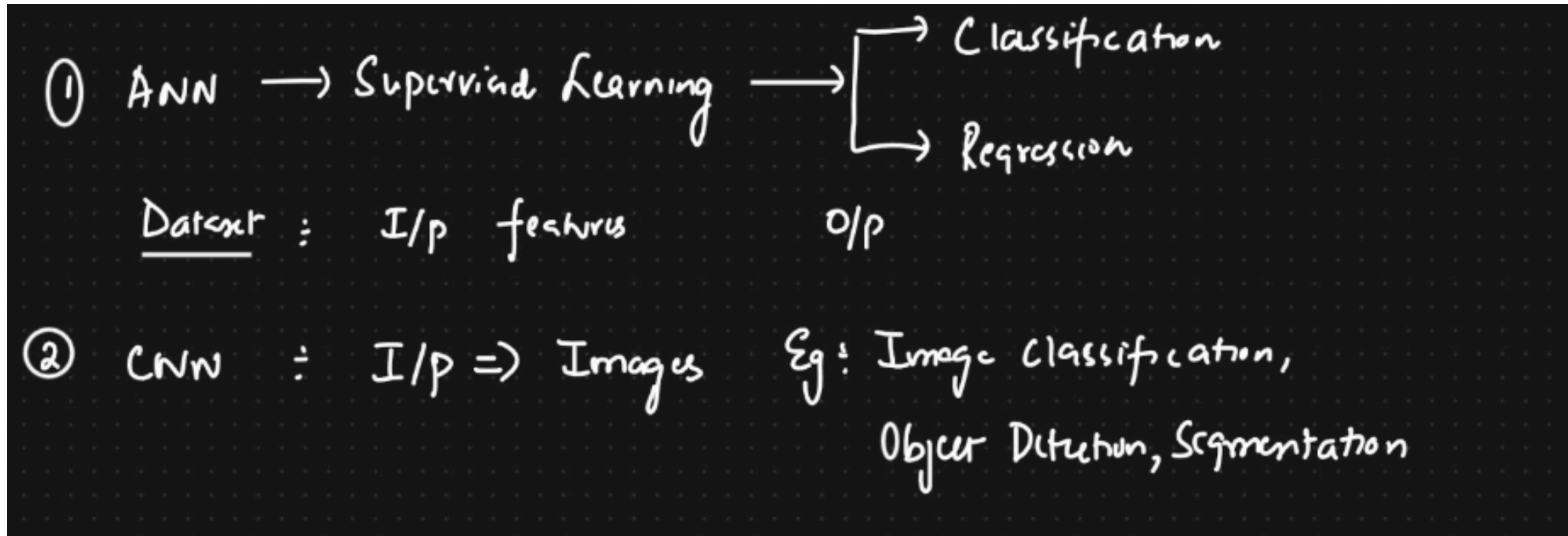


An Introduction to Convolutional Neural Networks (CNNs)



Why??

Because it's a part of your project:((



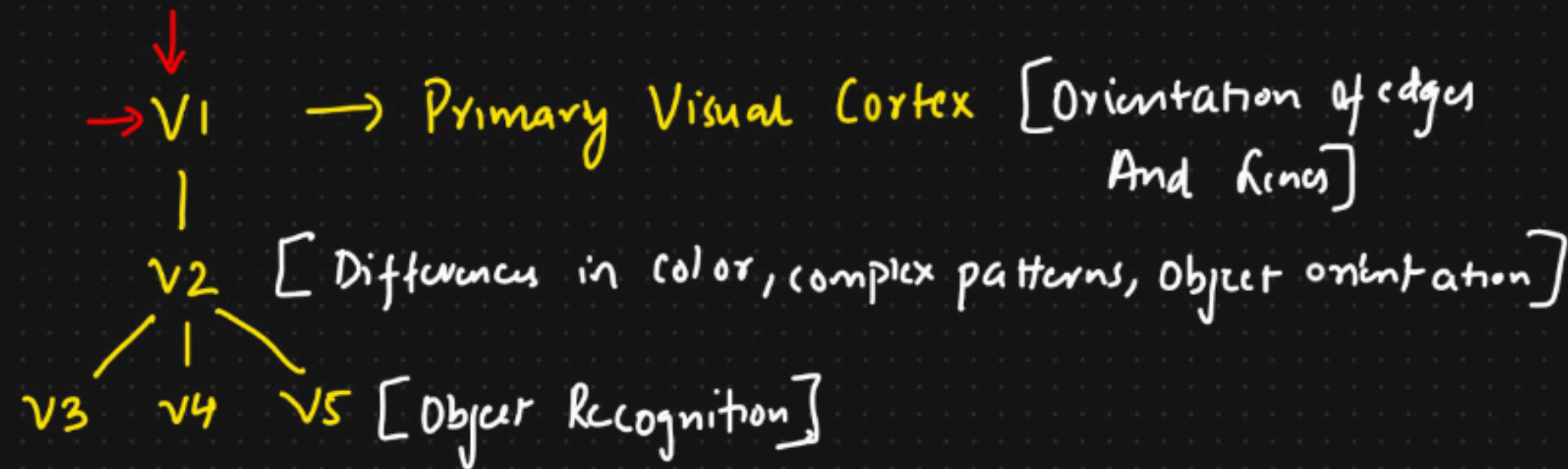
More Inspiring

The agenda for this field is to enable machines to view the world as humans do, perceive it in a similar manner, and even use the knowledge for a multitude of tasks such as Image & Video recognition, Image Analysis & Classification, Media Recreation, Recommendation Systems, Natural Language Processing, etc. The advancements in Computer Vision with Deep Learning have been constructed and perfected with time, primarily over one particular algorithm — a **Convolutional Neural Network**.

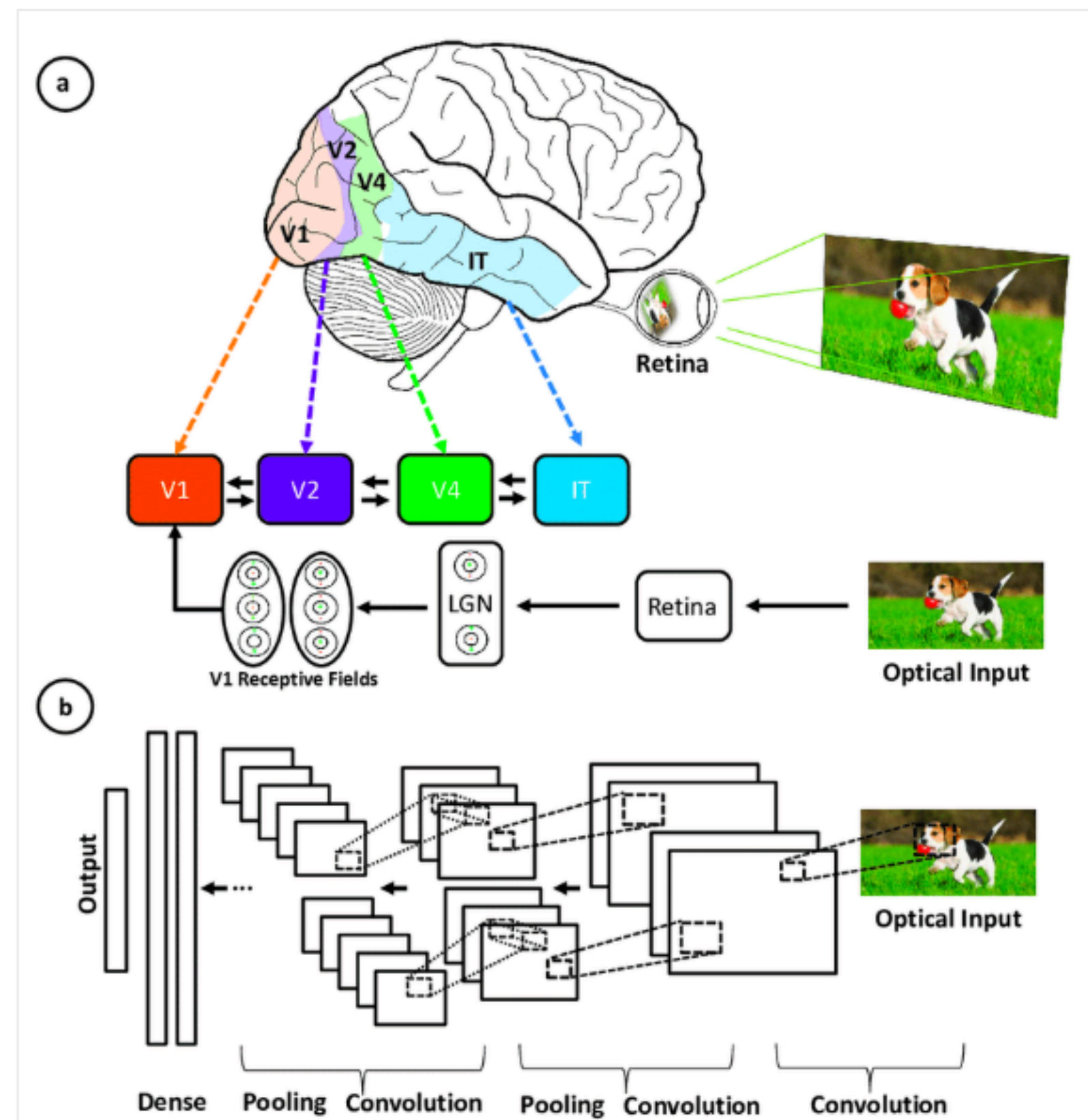
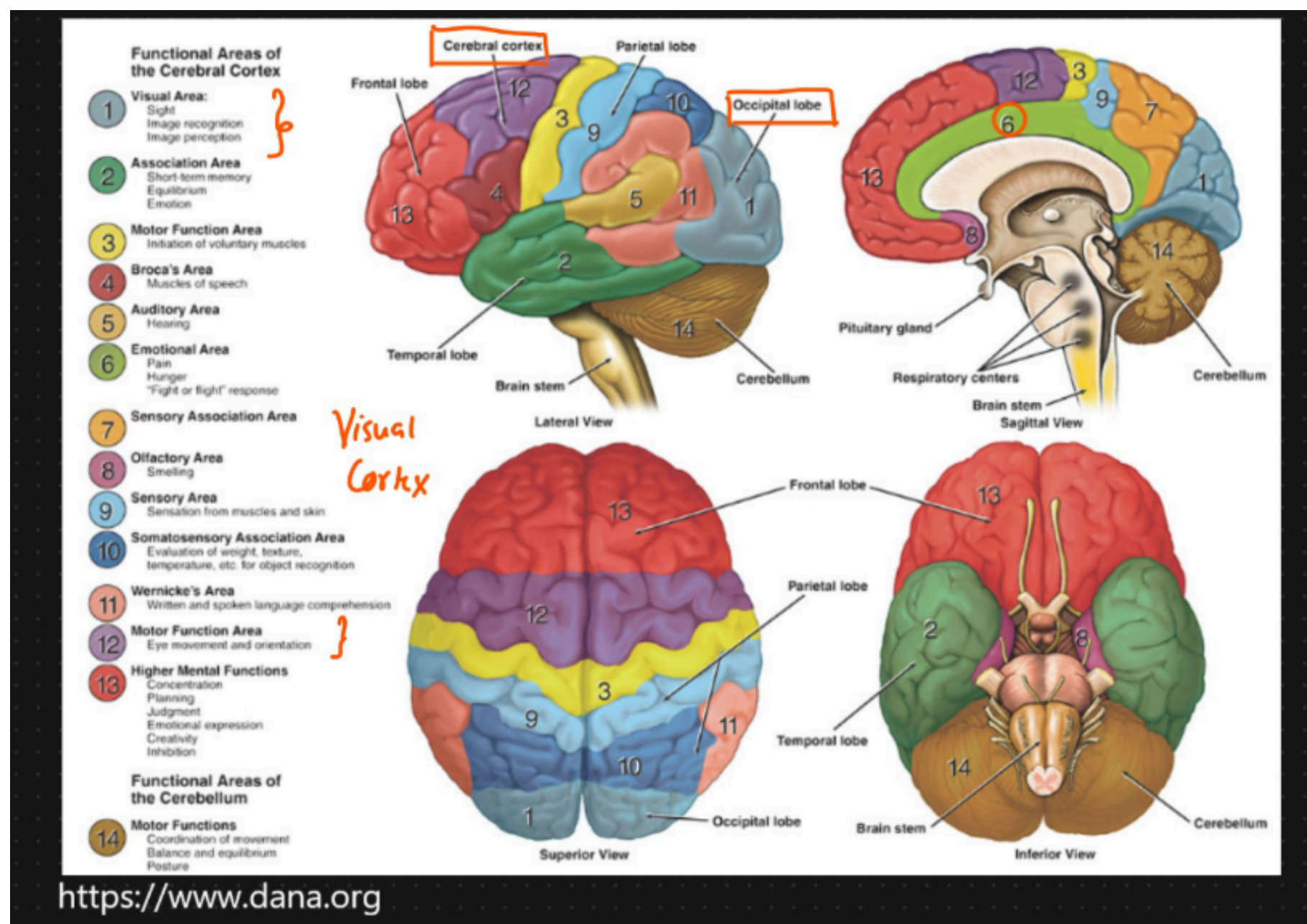
From where??

Convolutional neural networks were inspired by the layered architecture of the human visual cortex

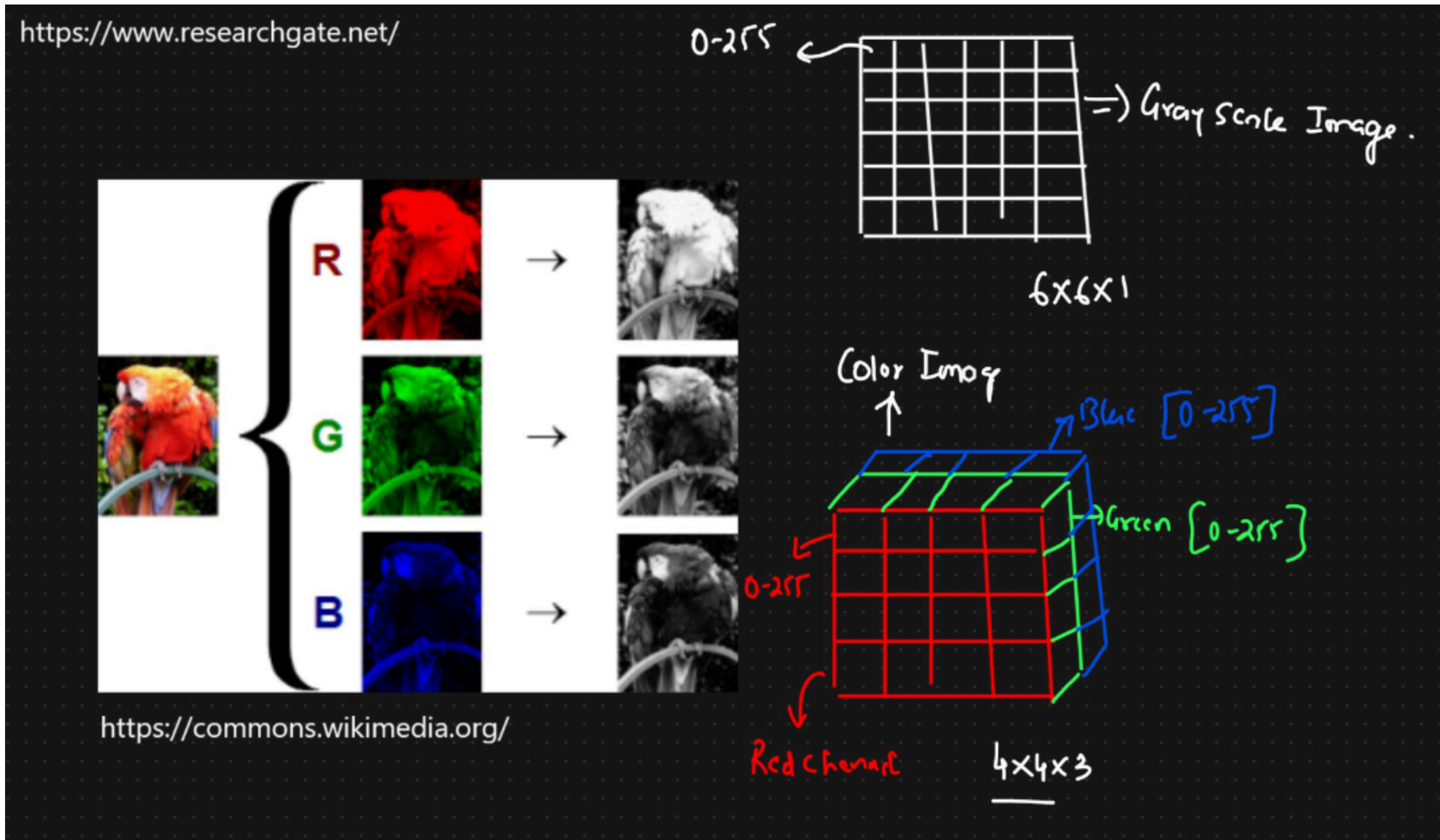
Visual Cortex (V1-V5) [Region of the brain that receives, integrates and processes visual information relayed from the retinas].



Visualize the Image



About images...

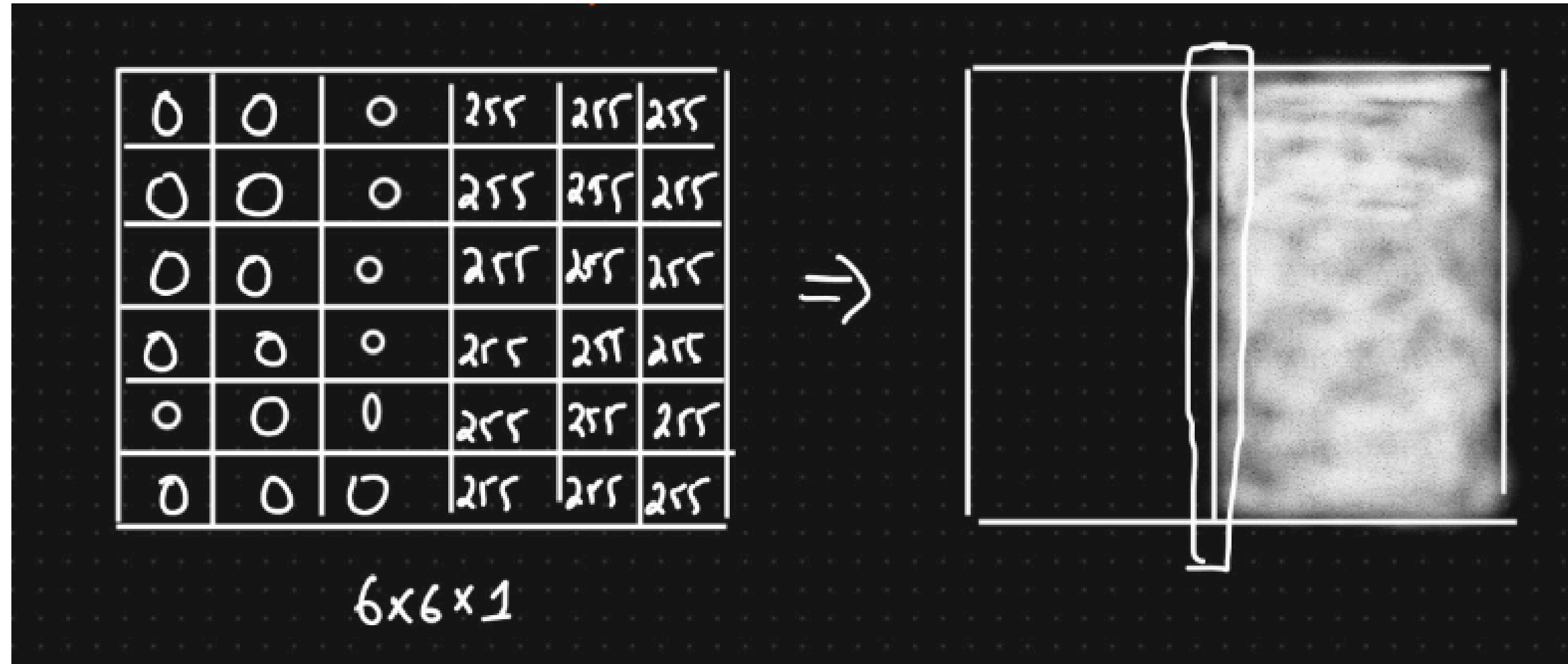


Let's start...

Key Components of a CNN

- Convolutional layers
- Rectified Linear Unit (ReLU for short)
- Pooling layers
- Fully connected layers

Convolution operation:



Firstly, We will normalise by Min-Max Scaling

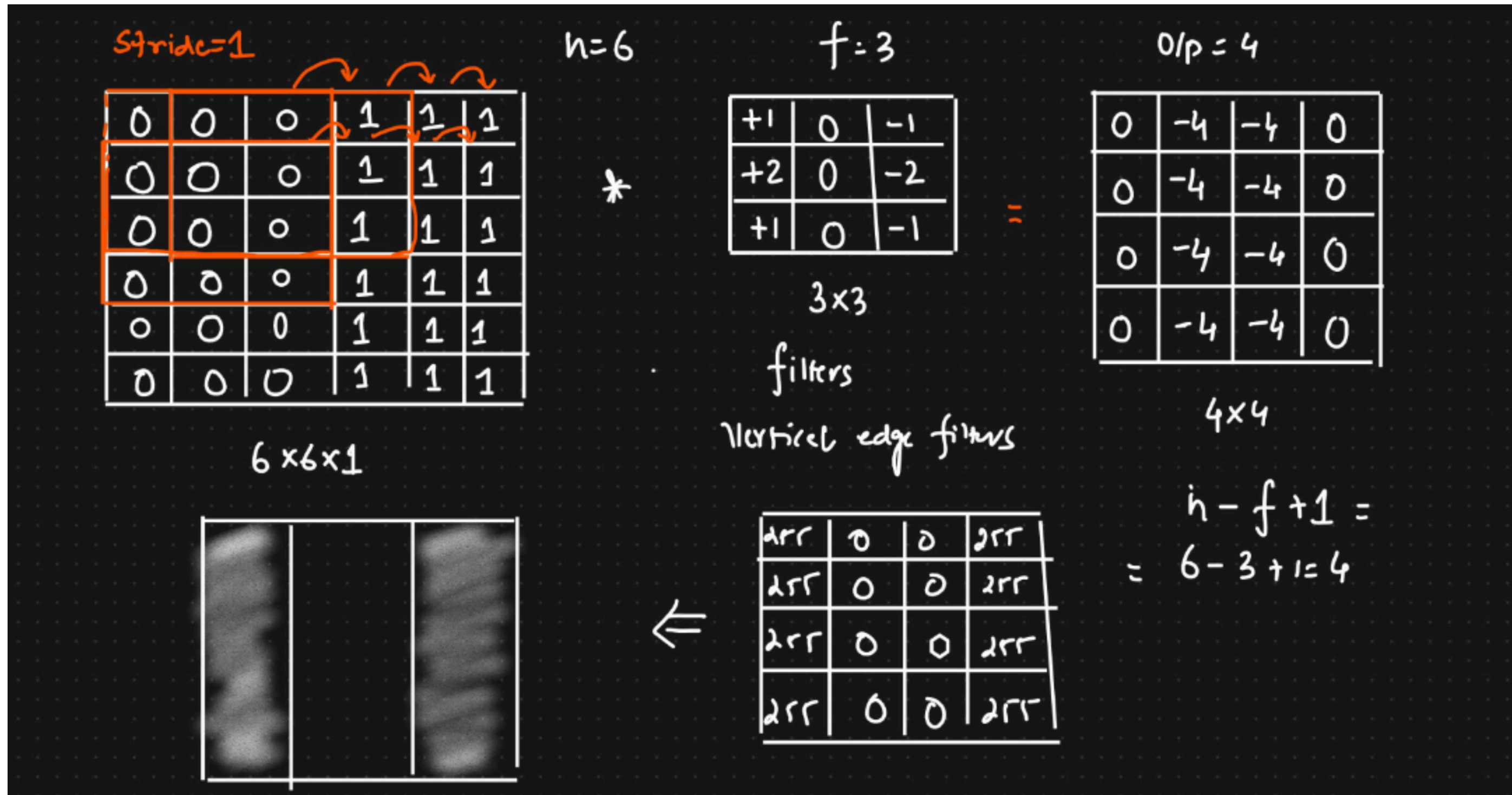
Min-Max Scaling

Min-max scaling is very often simply called 'normalization.' It transforms features to a specified range, typically between 0 and 1. The formula for min-max scaling is:

$$X_{\text{normalized}} = (X - X_{\text{min}}) / (X_{\text{max}} - X_{\text{min}})$$

$$X_{\text{min}} = 0$$

$$X_{\text{max}} = 255$$



Secondly, apply the filter/kernel by sliding window function to matrix.
 Third, revert back by feature scaling:
 min value=-4 ==> 0(min in range)
 max value=0 ==> 255(max in range)



Padding:

⑤ Padding In CNN

0	0	0	0	1	1	1	1
0	0	0	0	1	1	1	1
0	0	0	0	1	1	1	1
0	0	0	0	1	1	1	
0	0	0	0	1	1	1	
0	0	0	0	1	1	1	
0	0	0	0	1	1	1	
0							

*

- ① Zero padding
- ② Neighbour padding

$$f=3$$

+1	0	-1
+2	0	-2
+1	0	-1

=

	0	-4	-4	0	
	0	-4	-4	0	
	0	-4	-4	0	
	0	-4	-4	0	

6x6
=

6x6

$n=6$

8x8

$$n-f+2p+1$$

$$6-3+2p+1=6$$

$$3+2p+1=6$$

$$2p=6-4$$

$$p=\frac{2}{2}=1$$

To know no of
padding layers to be
added

Updated formula:
considering
strides(s)
 $(n-f+2p+1)/s$

If the size of
matrix is
decreasing, you
are losing
information.

ReLU

The filter need to be made as per the input image.

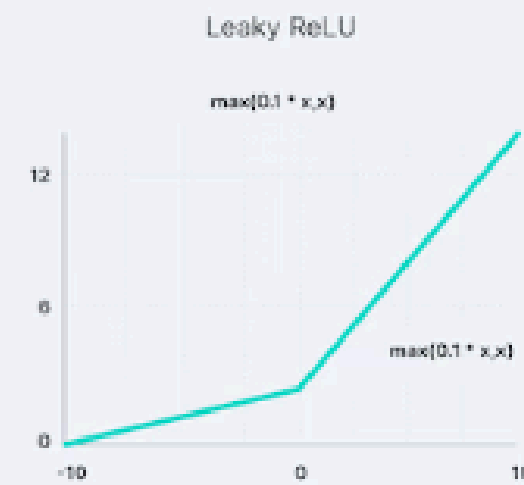
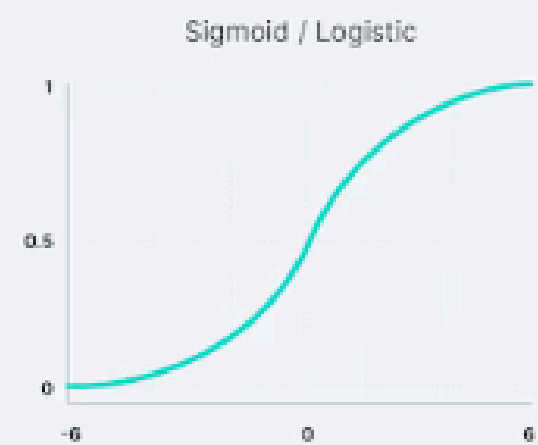
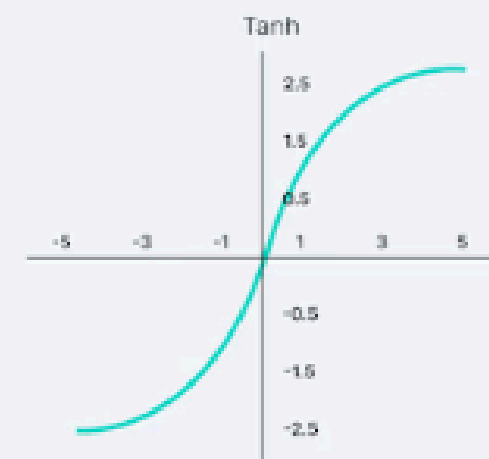
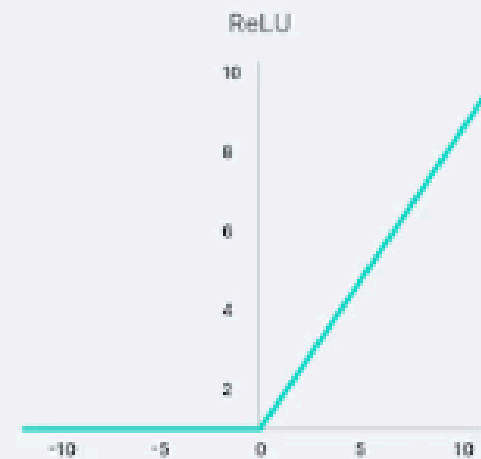
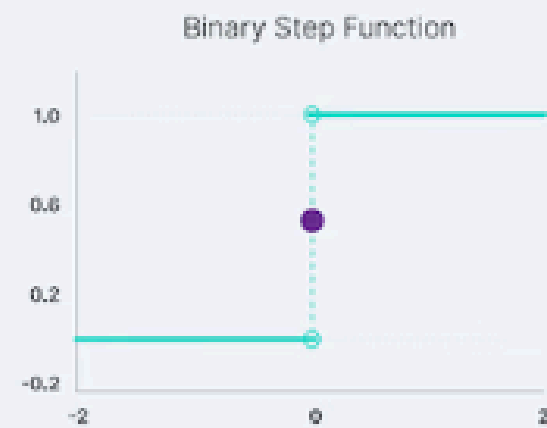
Backpropagation.

Activation Function.

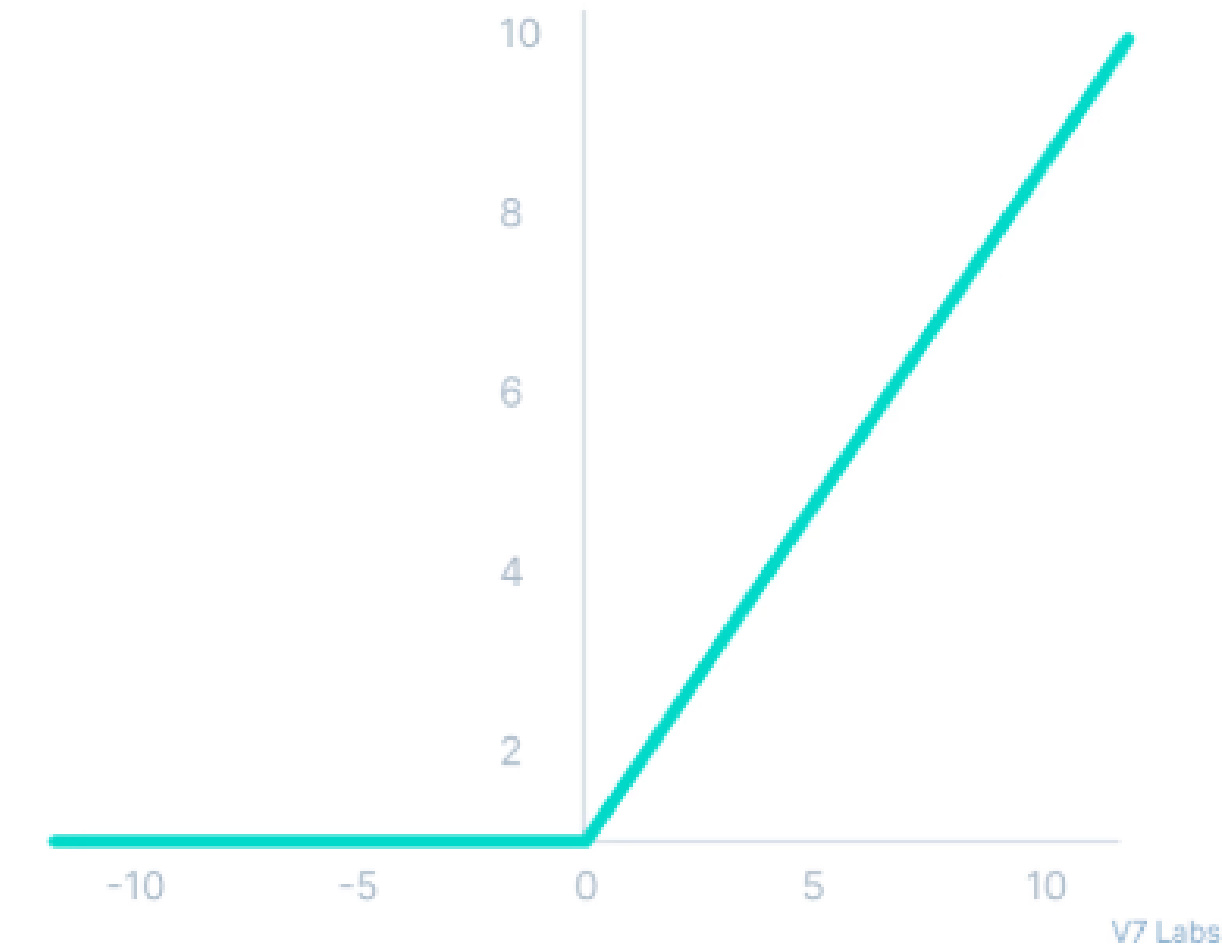
Most popular one is ReLU.

$$f(x) = \max(0, x)$$

Neural Network Activation Functions

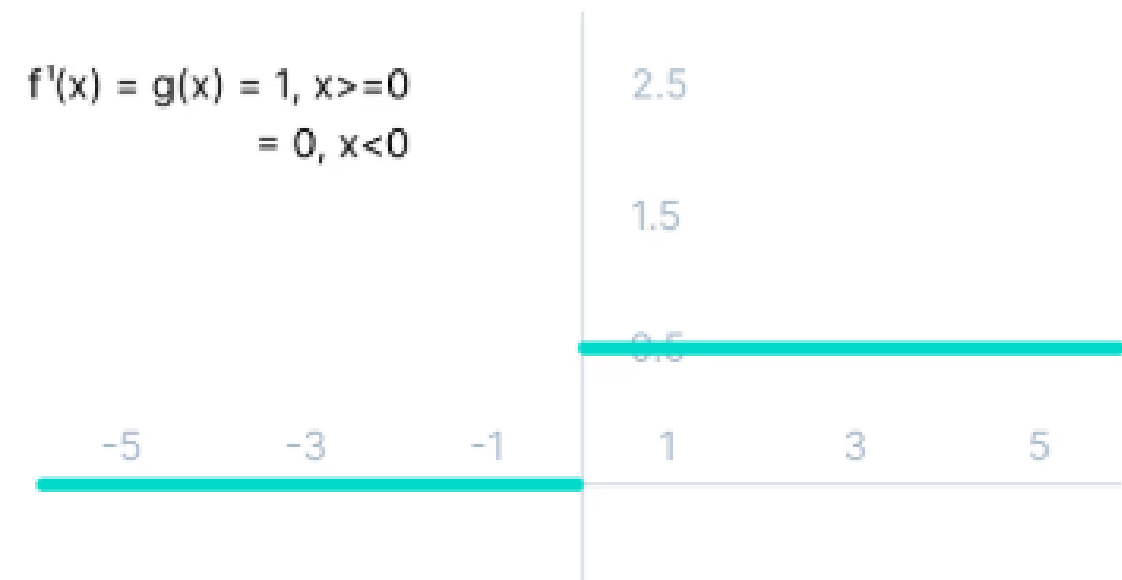


ReLU



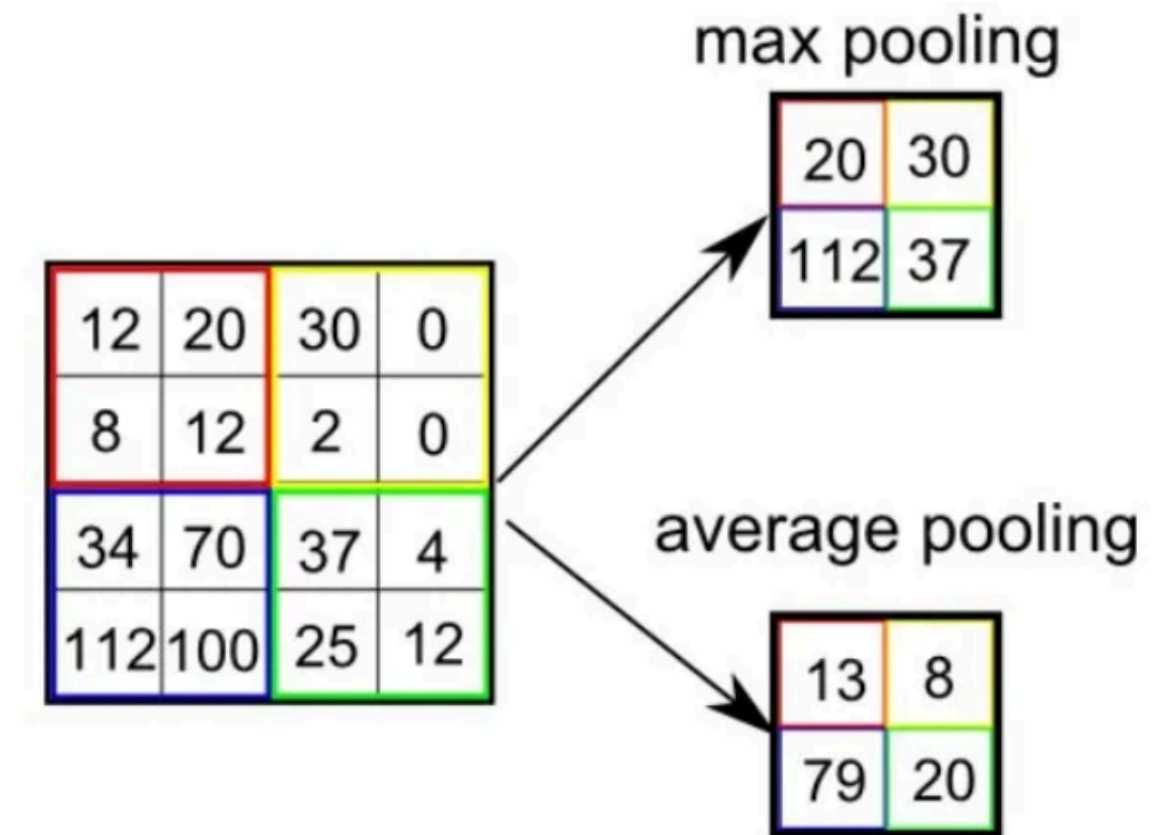
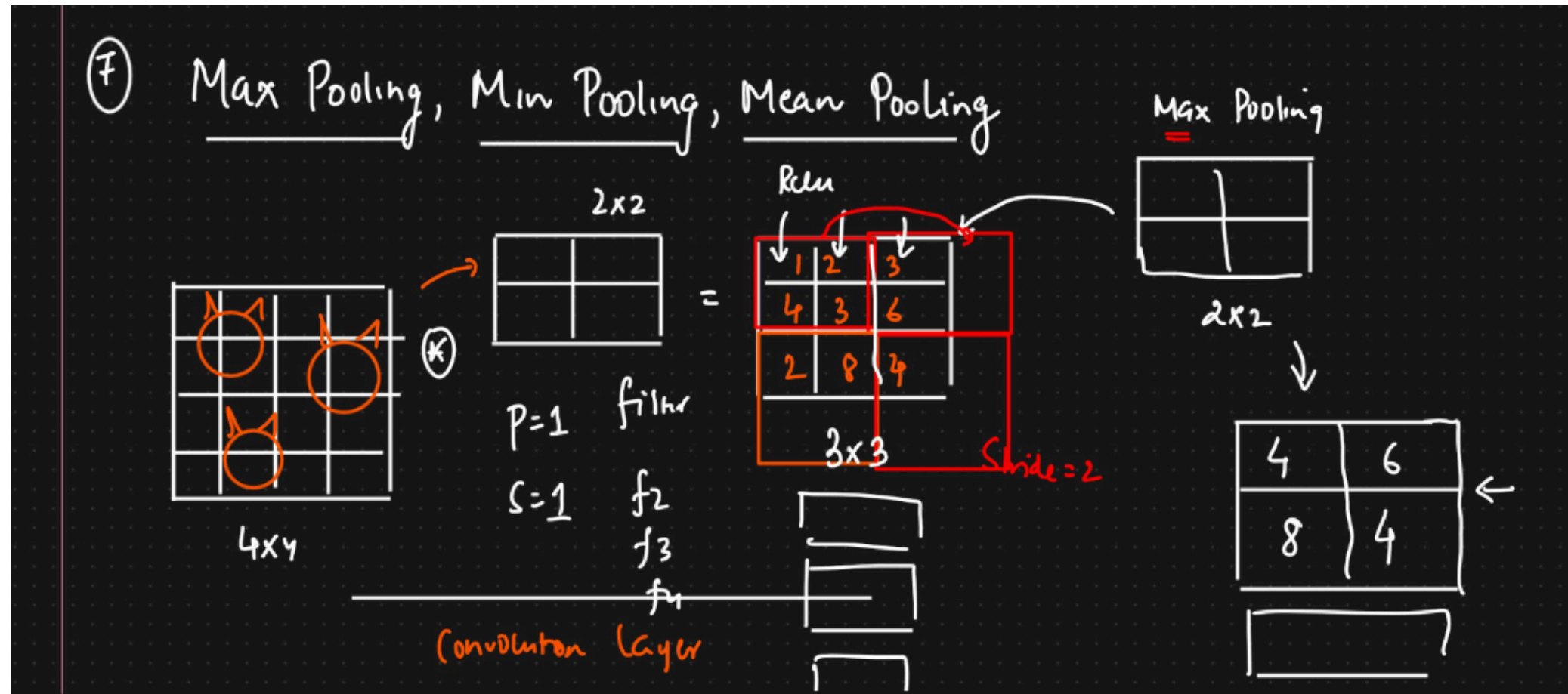
The Dying ReLU problem

$$f'(x) = g(x) = 1, x \geq 0 \\ = 0, x < 0$$



Pooling Layer

extracting dominant features

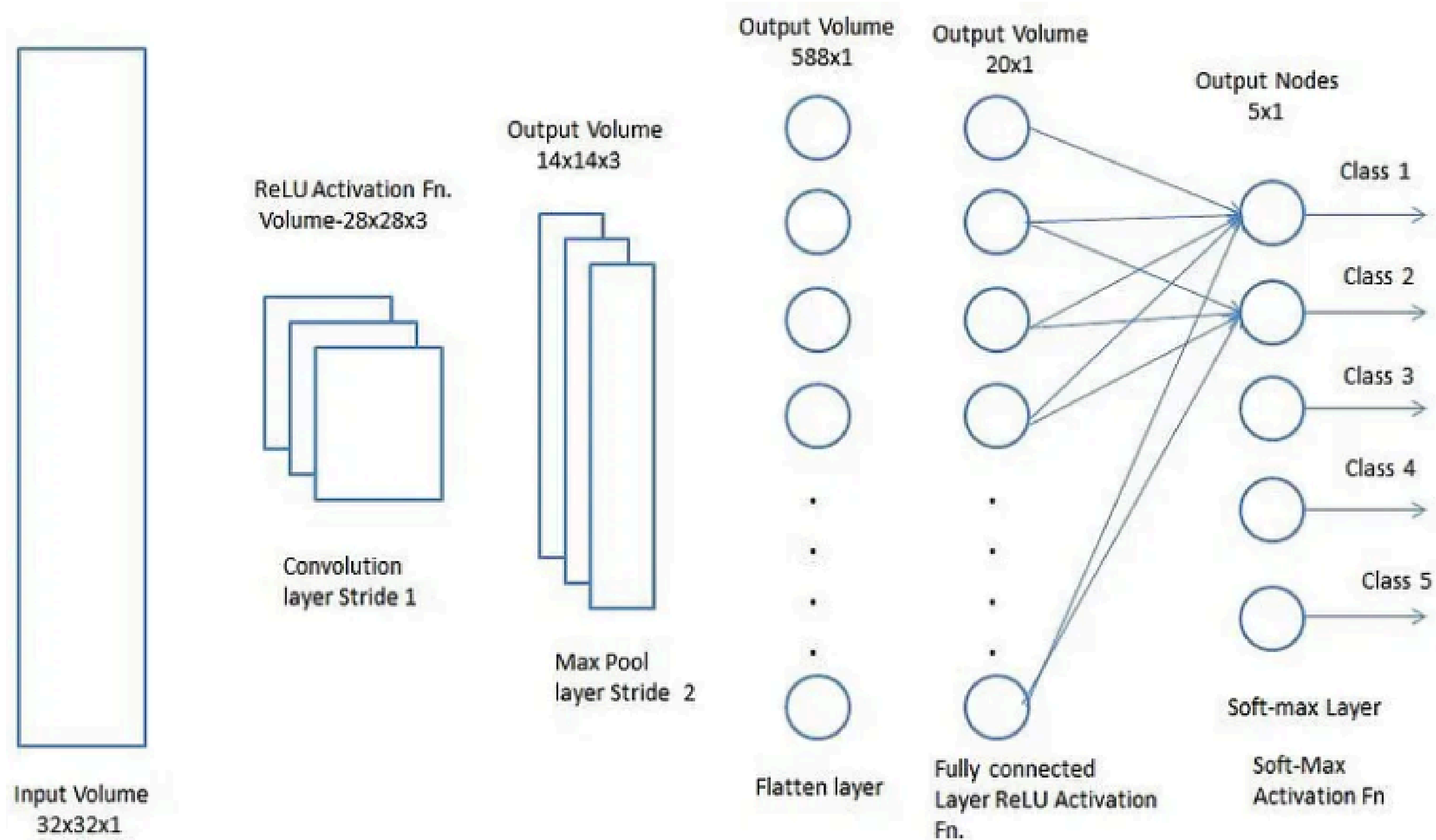


Types of Pooling

The dimension of the feature map becomes smaller as the pooling function is applied.

The last pooling layer flattens its feature map so that it can be processed by the fully connected layer.

Fully connected layers

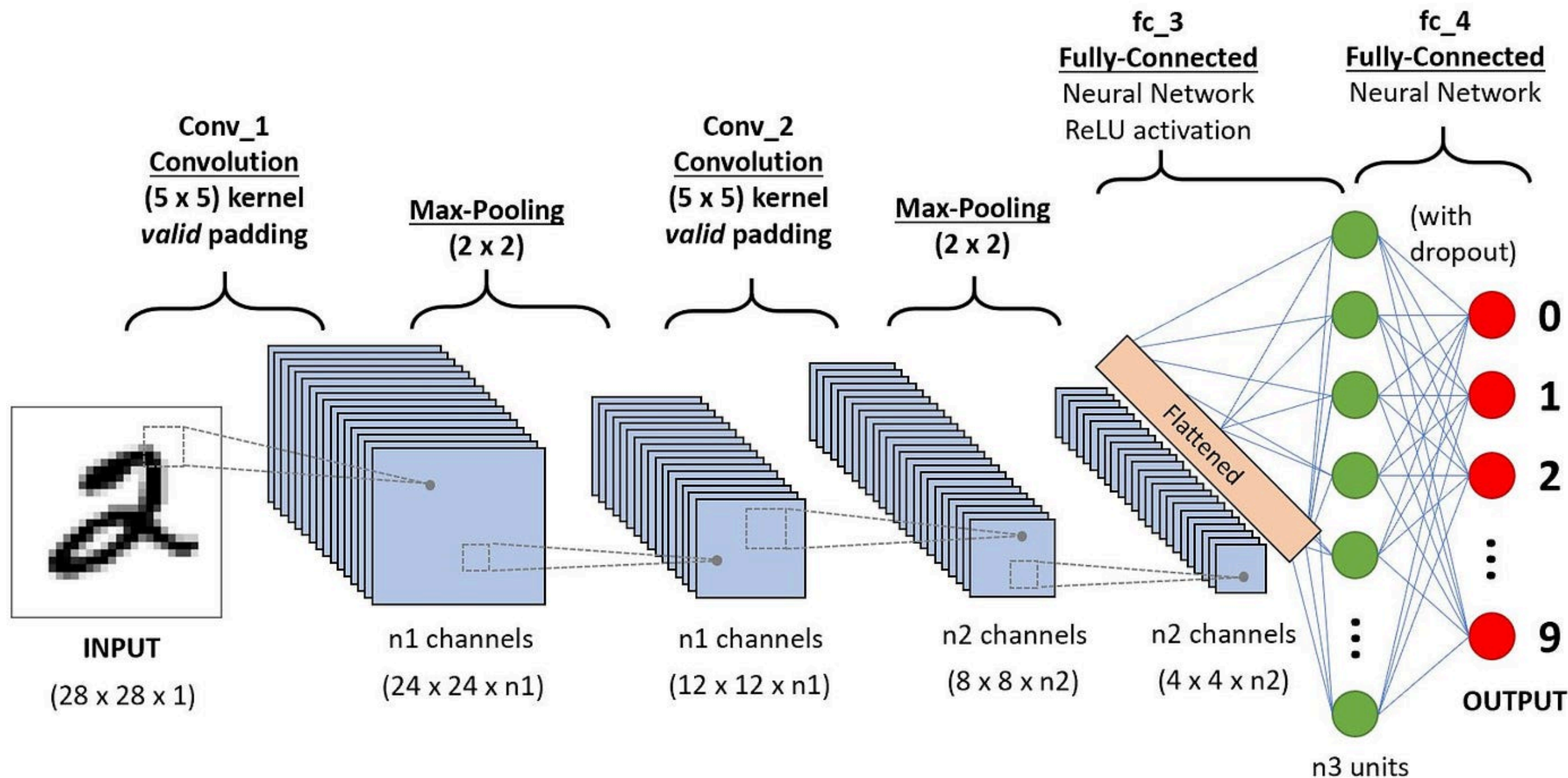


These layers are in the last layer of the convolutional neural network, and their inputs correspond to the flattened one-dimensional matrix generated by the last pooling layer.

What next?

Our input image is converted into a suitable form for our Multi-Level Perceptron, and flatten the image into a column vector.

The flattened output is fed to a feed-forward neural network and backpropagation is applied to every iteration of training.



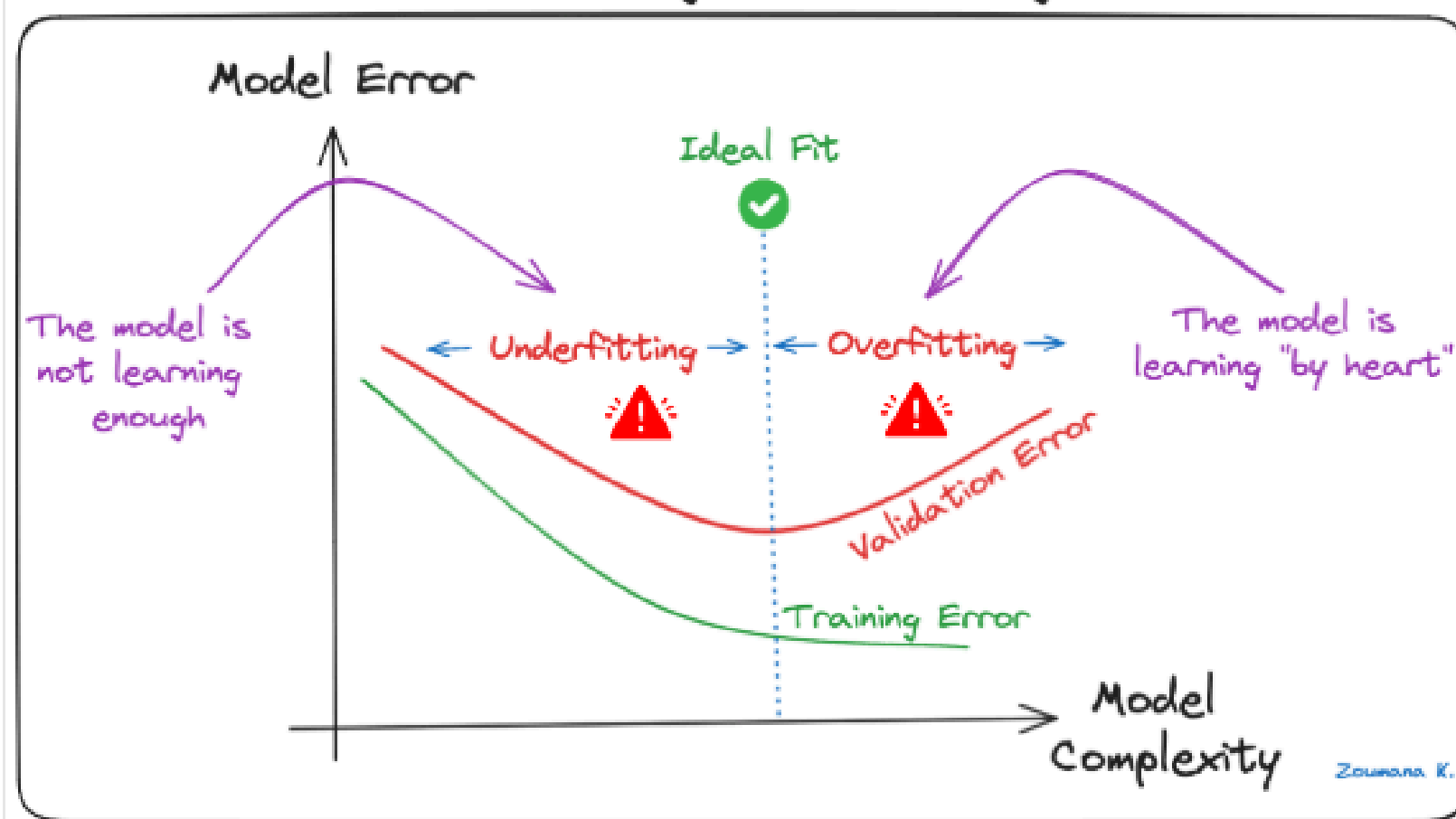
Over a series of epochs, the model is able to distinguish between dominating and certain low-level features in images and classify them



Google Colab

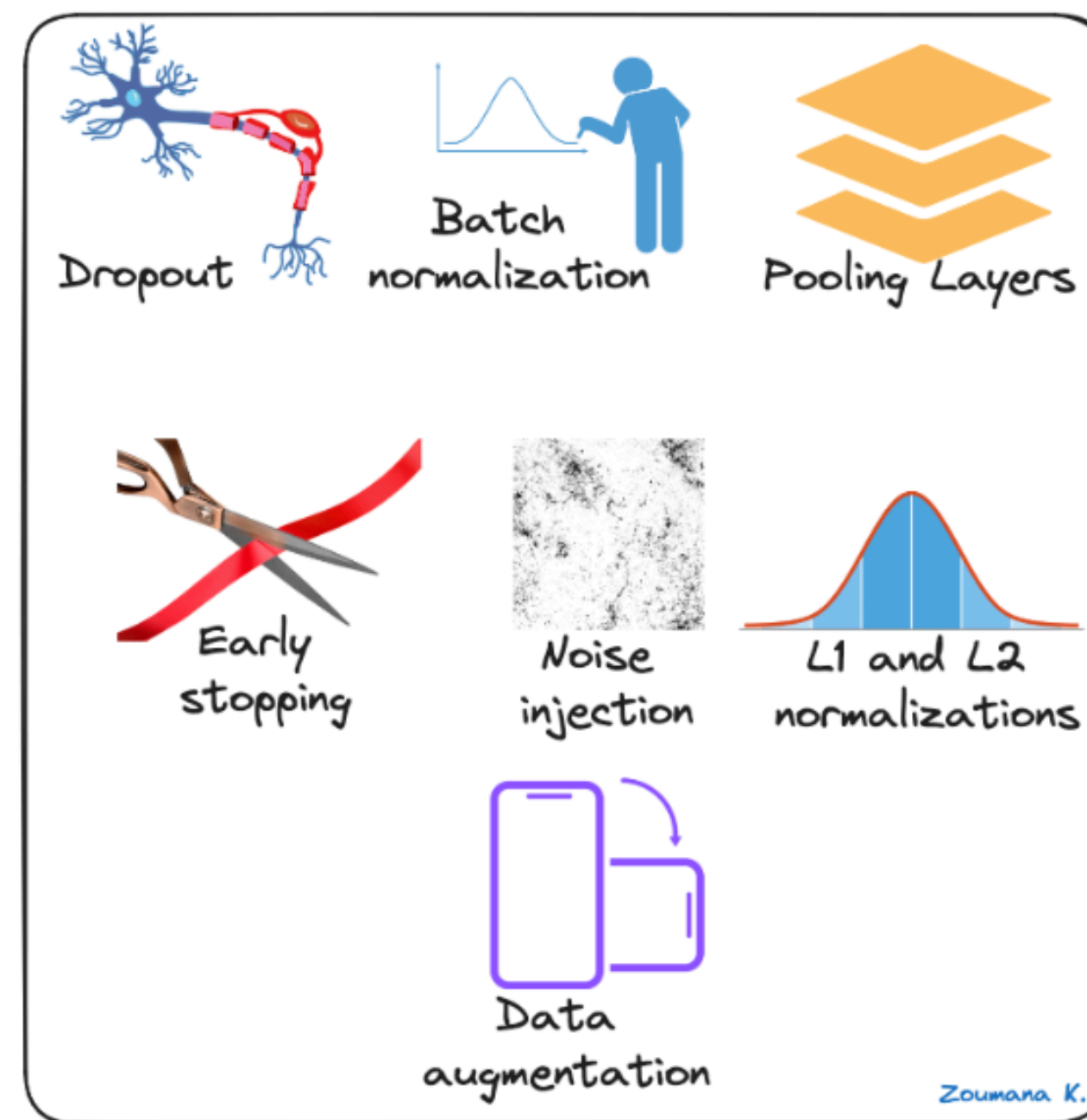
google.com

Underfitting Vs. Overfitting

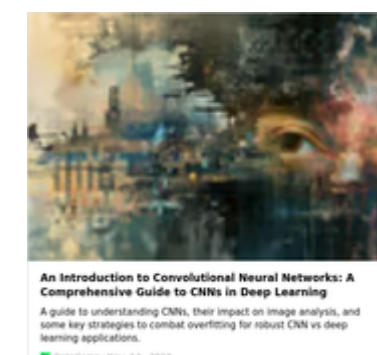


Underfitting Vs. Overfitting

7 Strategies to Mitigate Overfitting in CNNs



7 strategies to mitigate overfitting in CNNs



An Introduction to Convolutional Neural Networks: A Comprehensive Guide to CNNs in Deep Learning

A guide to understanding CNNs, their impact on image analysis, and some key strategies to combat overfitting for robust CNN vs deep learning applications.

datacamp Nov 14, 2023